



CONTACT

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LANGUAGE

Italian

English

SKILLS

Languages:

- C/C++
- Python
- GDScript

Frameworks:

- FastAPI

Operating Systems:

- Linux
- Windows
- macOS

Game Engines:

- Godot

Tools & Version Control:

- Git/GitHub
- Docker
- Makefile
- Shell
- CI/CD

HOBBIES

PC hardware

AI / Low levels performance trends

TOBIA CACCAVALE

C/C++ PYTHON DEVELOPER

Developer with a rigorous academic background in Medicine (Final year, 29.8/30 GPA) and intensive technical training at 42. Passionate about low-level optimization, bitwise operations, branchless programming, and performance-oriented programming. Proven track record of high-intensity commitment (12+ hours/day at 42) and rapid skill acquisition.

PROJECTS

MiniShell - C Team project

- Unix-like shell with complex parsing, pipe handling, redirections, and environment variable expansion.

SoLong - C Solo Project

- Created a comprehensive, feature-rich version of the project to earn entry into "Spaghetti Studio" (a competitive club for 42 game developers).
- Implemented a custom, real-time particle system generated from scratch without shaders.

Cub3D - C++ Team Project

- Rick & Morty-themed raycasting engine. Focused on rendering pipeline efficiency and cache-conscious memory access.

Transcendance - Python Team Project

- Architected the backend using FastAPI.
- Implemented a robust worker-based system handling independent game loops for each player and a high-frequency global game world loop for real-time synchronization.

Philosophers - C Solo Project

- Multithreaded simulation resolving concurrency, starvation, and mutual exclusion using mutexes and semaphores.

EXPERIENCE & EDUCATION

42 Network - Rome

- Completed core curriculum in ~1 year with an intense dedication. Focused on system-level architecture, performance, and algorithmic rigor.

Spaghetti Studio Member

- Selected for a competitive game-dev collective at 42 Network.

Academic Background

- Final year Medical Student (GPA 29.8/30). Developed high-level problem-solving, critical analysis, and extreme dedication to mastering high-stakes systems.
- High School Diploma in Scientific Studies (100/100).